PROJECT WORK REPORT

Data Analytics Projects: Supply Chain Analysis and Marketing Attribution Dashboard

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Role: Data Analyst / SQL Developer

Tools: Python, PostgreSQL, SQL, Power BI, Excel, Git, GitHub, Pandas, Matplotlib, Pytest

1. Introduction

This report presents the analytical and technical work completed across two major data analytics projects: **Supply Chain Analysis** and **Marketing Attribution Dashboard**.

The projects were developed to demonstrate practical data analytics capabilities across the complete data lifecycle, including data preparation, exploratory data analysis, database development, advanced SQL analysis, statistical calculations, business intelligence, attribution modelling, testing, documentation, and version control.

The Supply Chain Analysis project focused on understanding operational performance, inventory behaviour, sales trends, product performance, customer activity, and supply chain efficiency.

The Marketing Attribution Dashboard project focused on analysing customer journeys, marketing interactions, conversions, revenue contribution, campaign performance, and return on marketing investment using multiple attribution models.

Together, the projects demonstrate the ability to move from **raw data to structured analytical insights and interactive business reporting**.

2. Project One: Supply Chain Analysis

2.1 Project Objective

The objective of the Supply Chain Analysis project was to analyse supply chain and sales data to identify patterns in product demand, inventory performance, sales activity, customer behaviour, and operational efficiency.

The project was designed to answer business questions such as:

- Which products generate the highest revenue?
- Which products have the highest demand?
- Which locations or categories perform best?
- How does sales performance change over time?
- Which products may require closer inventory monitoring?
- What factors contribute to supply chain performance?
- How can historical data support better business decisions?

2.2 Data Preparation

The project involved preparing raw datasets for analysis.

The data preparation process included:

1. Importing raw datasets.
2. Inspecting the structure and variables.
3. Identifying missing values.
4. Checking duplicate records.
5. Standardising column names.
6. Converting dates into appropriate formats.
7. Cleaning numerical fields.
8. Checking categorical variables.
9. Creating analytical variables.
10. Preparing cleaned datasets for visualization and reporting.

Python and Pandas were used extensively for data manipulation and exploratory analysis.

2.3 Exploratory Data Analysis

Exploratory Data Analysis was conducted to understand the underlying characteristics of the dataset.

The analysis examined:

- Sales distribution
- Product performance
- Revenue trends
- Quantity sold
- Customer activity
- Category performance
- Geographic or location performance
- Temporal patterns
- Potential outliers
- Missing data patterns

Descriptive statistics and visualizations were used to identify important trends before proceeding to deeper analysis.

2.4 Business Performance Analysis

The analysis moved beyond simple descriptive statistics to evaluate business performance.

Important measures included:

- Total revenue
- Total sales
- Units sold
- Average order value
- Product contribution
- Category contribution
- Sales growth
- Customer activity
- Inventory-related indicators

These metrics provided a structured view of operational and commercial performance.

2.5 Data Visualization

Visualizations were developed to communicate analytical findings clearly.

The visual analysis included charts for:

- Sales trends
- Product performance
- Category comparison
- Revenue distribution
- Customer activity
- Geographic performance
- Key operational indicators

The purpose of the visualizations was not only to display data but to transform analytical results into information that could support managerial decision making.

2.6 Dashboard Development

The analytical results were prepared for dashboard presentation.

The dashboard approach focused on presenting important KPIs and trends in an easily interpretable format.

The dashboard can be used to monitor:

Revenue → Sales → Products → Customers → Operational Performance

This provides a high-level view while allowing individual business areas to be investigated in greater detail.

3. Project Two: Marketing Attribution Dashboard

3.1 Project Objective

The Marketing Attribution Dashboard was developed to analyse how different marketing channels and campaigns contribute to customer conversions and revenue.

The project uses customer journey data to determine how marketing interactions contribute to a conversion rather than attributing the entire conversion to a single marketing channel.

The project therefore combines:

Data ingestion → Data cleaning → Customer journeys → Attribution modelling → KPI analysis → Visualization

4. Marketing Data Pipeline

A structured analytical pipeline was established.

The project structure includes:

```
data/
├── raw/
└── processed/

docs/

notebooks/

sql/

src/
```

├── analytics/

├── attribution/

├── config/

├── ingestion/

├── pipeline/

├── preprocessing/

├── utils/

└── visualization/

tests/

dashboards/

This structure separates raw data, processing logic, analytical models, SQL operations, testing, documentation and visualization.

5. Data Ingestion

The ingestion layer was developed to load marketing datasets into the analytical workflow.

The ingestion components include:

data_loader.py

load_csv.py

load_data.py

validate_schema.py

The purpose of this layer is to provide a controlled process for bringing raw marketing data into the project.

Schema validation was also incorporated to ensure that incoming datasets contain the expected fields.

6. Data Preprocessing

The preprocessing layer was developed to prepare raw marketing data for analysis.

The work includes:

- Missing-value handling

- Duplicate removal
- Date transformation
- Data cleaning
- Feature engineering
- Revenue normalization
- Dataset preparation

The preprocessing workflow ensures that downstream SQL, Python and attribution calculations operate on consistent data.

7. Customer Journey Construction

One of the major components of the project is customer journey analysis.

Marketing interactions are organised according to:

Customer → Interaction → Channel → Campaign → Conversion

For example:

Customer

↓

Facebook

↓

Google

↓

Email

↓

Conversion

This allows the project to examine the sequence of interactions that occurs before a customer converts.

8. Attribution Modelling

Several attribution models were implemented.

First Touch Attribution

First Touch Attribution assigns conversion credit to the customer's first eligible marketing interaction.

First interaction → 100% credit

This model is useful for evaluating channels responsible for initiating customer awareness.

Last Touch Attribution

Last Touch Attribution assigns conversion credit to the final eligible interaction before conversion.

Last interaction → 100% credit

This helps identify channels that are effective at driving customers toward conversion.

Linear Attribution

Linear Attribution distributes conversion credit equally across all eligible touchpoints.

For example:

Facebook → 33.3%

Google → 33.3%

Email → 33.3%

This provides a more balanced representation of multi-touch customer journeys.

Position-Based Attribution

A U-shaped position-based model was implemented.

The model gives greater importance to the first and last interactions while distributing the remaining credit across intermediate interactions.

For example, with three touchpoints:

First	Middle	Last
40%	20%	40%

This model recognises both customer acquisition and conversion-driving interactions.

Time Decay Attribution

Time Decay Attribution was also incorporated into the project's advanced attribution work.

The model assigns greater attribution weight to interactions closer to the conversion event.

Conceptually:

Older interaction → Lower weight

Recent interaction → Higher weight

This allows the analysis to account for the possibility that recent marketing interactions have a stronger influence on conversion behaviour.

9. Advanced SQL Development

A major component of Isaac's contribution to the Marketing Attribution Dashboard is the SQL and database development layer.

The project goes beyond basic SELECT, GROUP BY and filtering queries.

Advanced SQL analysis includes:

Monthly Trend Analysis

Marketing performance is aggregated by month to identify changes in:

- Revenue
- Conversions
- Marketing spend
- Attributed revenue
- Return on advertising spend

Weekly Trend Analysis

Weekly aggregation allows shorter-term campaign and marketing performance changes to be identified.

Stored Procedures

Stored procedures are being developed to encapsulate recurring analytical operations and allow database-side execution of business logic.

Time Decay Attribution

SQL calculations are used to assign conversion weights according to the time difference between an interaction and conversion.

Position-Based Attribution

SQL logic assigns U-shaped attribution weights based on the position of each touchpoint in the customer journey.

Additional advanced SQL functionality includes:

- Common Table Expressions
- Window functions

- Conditional aggregation
- Date functions
- Ranking functions
- Customer journey sequencing
- Revenue allocation
- Campaign-level aggregation
- Channel-level performance analysis
- KPI calculations
- Analytical views

10. KPI Development

The project contains a dedicated analytics layer for calculating marketing performance indicators.

Important KPIs include:

Customer Acquisition Cost

Measures the cost associated with acquiring customers.

Return on Advertising Spend

Measures revenue generated relative to advertising expenditure.

Return on Investment

Evaluates the financial return generated from marketing investment.

Conversion Rate

Measures the proportion of users or visitors that complete a desired conversion.

Attributed Revenue

Measures revenue assigned to marketing channels or campaigns according to an attribution model.

Customer and Campaign Performance

The dashboard also enables comparison of campaigns and channels according to conversion and revenue contribution.

11. Python Analytics

Python was used to support the analytical workflow.

The project contains analytical modules including:

cac.py

conversation_date.py

kpi_calculator.py

eda.py

roas.py

roi.py

These modules separate individual analytical responsibilities rather than placing all calculations in one script.

This makes the project easier to maintain, test and extend.

12. Visualization Development

A visualization layer was established to transform analytical outputs into charts.

The project includes components for:

charts.py

plot_campaigns.py

plot_channels.py

plot_roi.py

These components support visualization of:

- Campaign performance
- Channel performance
- ROI
- Revenue
- Marketing efficiency
- Attribution results

The final objective is to make the analytical results suitable for presentation through an interactive dashboard.

13. Power BI Dashboard

A Power BI dashboard file is included in:

dashboards/

└── marketing_attribution.pbix

The dashboard provides a business-facing presentation layer for the analytical results.

The intended analytical flow is:

Marketing Data

↓

Data Cleaning

↓

PostgreSQL

↓

Advanced SQL

↓

Attribution Models

↓

KPI Calculations

↓

Power BI

↓

Business Insights

This architecture demonstrates how SQL and Python analytical processing can feed a business intelligence environment.

14. Testing and Quality Assurance

Testing was incorporated into the project to improve reliability.

The project contains tests covering:

tests/

├── attribution/

├── analytics/

├── database/

├── ingestion/

├── pipeline/

├── preprocessing/

└── runner/

Attribution tests cover:

- First touch
- Last touch
- Linear attribution
- Position-based attribution
- Time decay attribution
- Attribution engine

Other tests cover:

- Data ingestion
- Data cleaning
- Date transformation
- Feature engineering
- Pipeline execution
- Analytics
- Database functionality

A custom testing framework was also developed under:

tests/runner/

This includes:

discovery.py

execution_handler.py

executer.py

mapper.py

reporter.py

test_generation.py

test_runner.py

The runner provides functionality for discovering, executing and reporting project tests.

15. Documentation

The project documentation layer was expanded to document the analytical and technical work.

Important documentation areas include:

- Project charter
- Database design
- KPI definitions
- API documentation
- Testing reports
- Final project reports
- Project progress
- Implementation documentation

The documentation provides a record of the project's objectives, methodology, technical architecture, analytical decisions and results.

16. Version Control and GitHub

Git was used to maintain the development history of the project.

The project follows a structured workflow:

Development

↓

Testing

↓

Git Add

↓

Git Commit

↓

Git Push

Work was organised into multiple commits to provide evidence of continuous development.

The GitHub repository serves as the central location for the source code, documentation, SQL scripts, tests and project structure.

17. Technical Skills Demonstrated

The two projects demonstrate practical experience in:

Area	Skills Demonstrated
Data Analysis	Exploratory and descriptive analysis
Python	Pandas, data processing, analytical scripting
SQL	Advanced querying, aggregation and attribution
PostgreSQL	Database design and analytical querying
Power BI	Dashboard development
Data Visualization	Analytical charts and KPI reporting
Marketing Analytics	Attribution and campaign analysis
Supply Chain Analytics	Product, sales and operational analysis
Testing	Pytest and custom test execution
Git	Version control and structured commits
Documentation	Technical and analytical reporting

18. Overall Project Outcome

The combined work demonstrates progression from **basic data analysis to advanced analytics engineering**.

The Supply Chain Analysis established practical experience in cleaning, exploring, analysing and visualising operational data.

The Marketing Attribution Dashboard extends these capabilities into a more advanced analytical engineering project involving:

PostgreSQL + Advanced SQL + Python + Attribution Modelling + KPI Engineering + Testing + Power BI + Git

The Marketing Attribution project particularly demonstrates the ability to implement business concepts such as customer journeys, marketing attribution, ROI and campaign performance using reproducible analytical workflows.

19. Conclusion

The completed projects provide practical evidence of the ability to work across different stages of a modern data analytics workflow. The Supply Chain Analysis focused on operational and commercial insights, while the Marketing Attribution Dashboard addressed more advanced marketing analytics and attribution problems.

The work completed so far establishes a strong technical foundation consisting of data ingestion, preprocessing, SQL analysis, Python analytics, attribution modelling, KPI development, testing, visualization and dashboard preparation.

The next stage of development is to further integrate the PostgreSQL database, complete the advanced SQL layer, connect analytical outputs to the dashboard, validate the complete pipeline and document the final business insights.